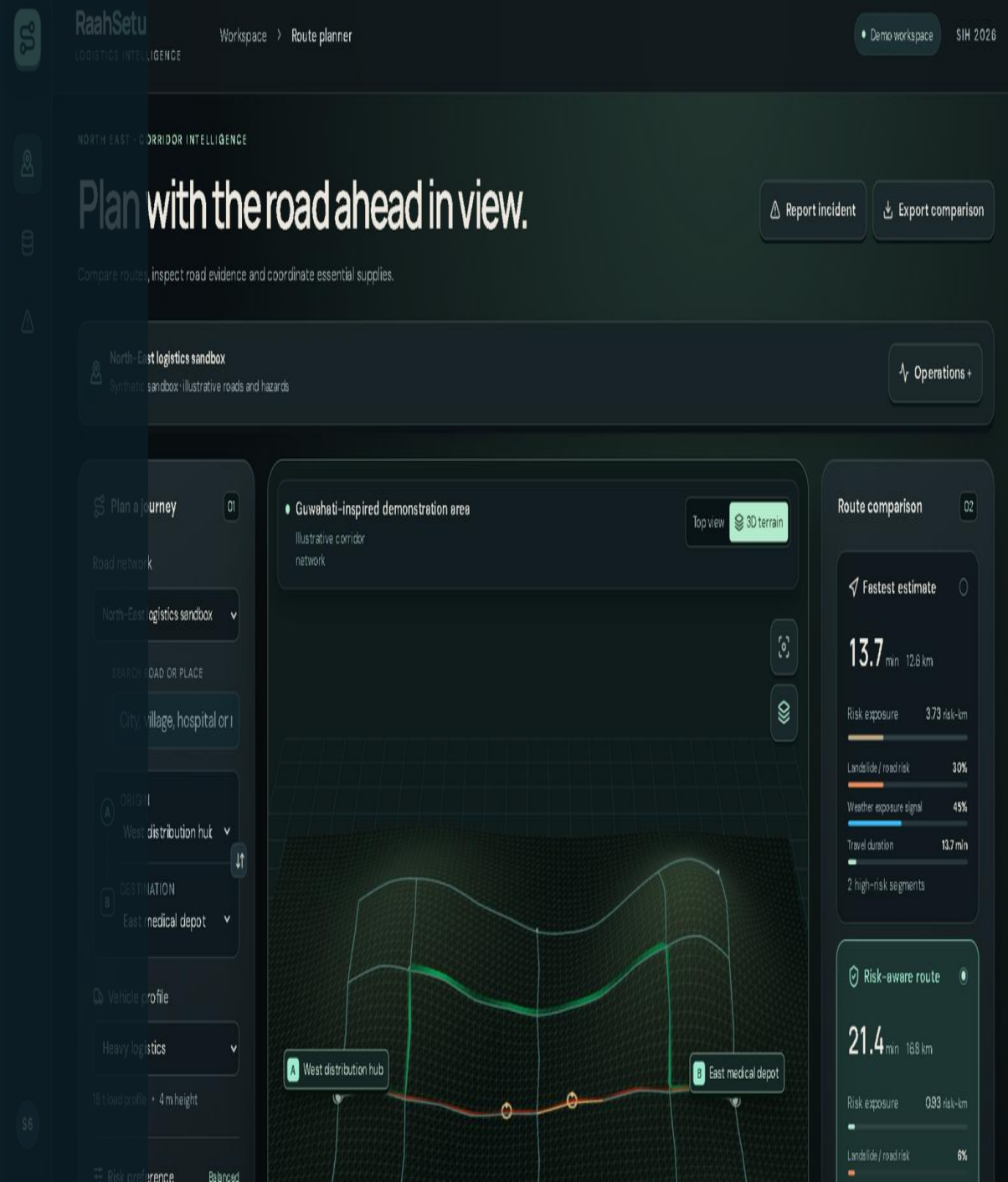


RaahSetu

Explainable risk-aware logistics routing for North-East India

Problem Statement ID	SIH26002
Problem Statement Title	Navigation and route optimization for freight and logistics in North-East India
Theme	Smart Automation / Logistics
PS Category	Software
Team ID	SIH-TBD (fill from portal)
Team Name	RaahSetu (as registered on portal)



Proposed Solution

A working prototype that plans heavy-vehicle journeys around road risks and shows why.

Two routes, one decision

Every journey returns the fastest route and a risk-aware alternative with time cost, distance and exposure side by side.

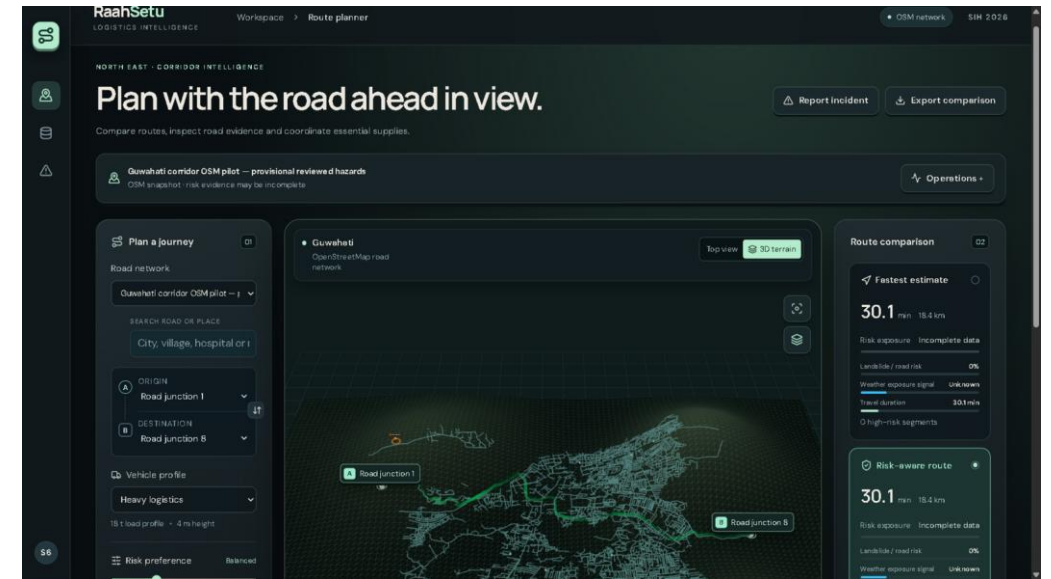
Risk with provenance

Accident history, road condition, weather and closures per road segment. Missing evidence stays “unknown” — never silently safe.

Field evidence loop

Officials report incidents with photo evidence offline; reviewers accept or reject before any road is treated as blocked.

75% lower modelled risk exposure on the demo corridor for 7.7 extra minutes — the tradeoff, quantified.



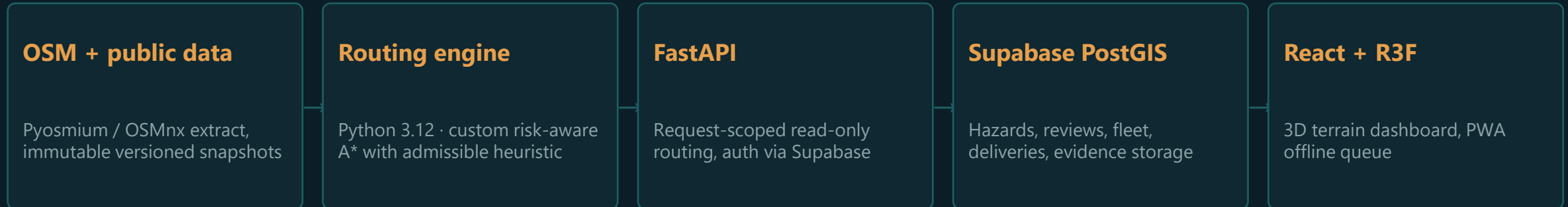
Live prototype — Guwahati corridor on real OSM roads

5,814 nodes · 13,681 directed segments · reviewed hazard observations

All-Northeast coverage prepared: 8 state OSM extracts, 273,494 road ways acquired

Technical Approach

Blueprint stack, fully working end-to-end on live Supabase.



The innovation: risk lives inside the edge cost

$$c(e) = t(e) \times [1 + 4\lambda \cdot r(e)] \quad r(e) = 0.5 \cdot \text{accident} + 0.3 \cdot \text{surface} + 0.2 \cdot \text{weather}$$

λ (risk preference) is the operator's dial — the same admissible A* stays provably optimal for each setting.

44

automated tests passing

300

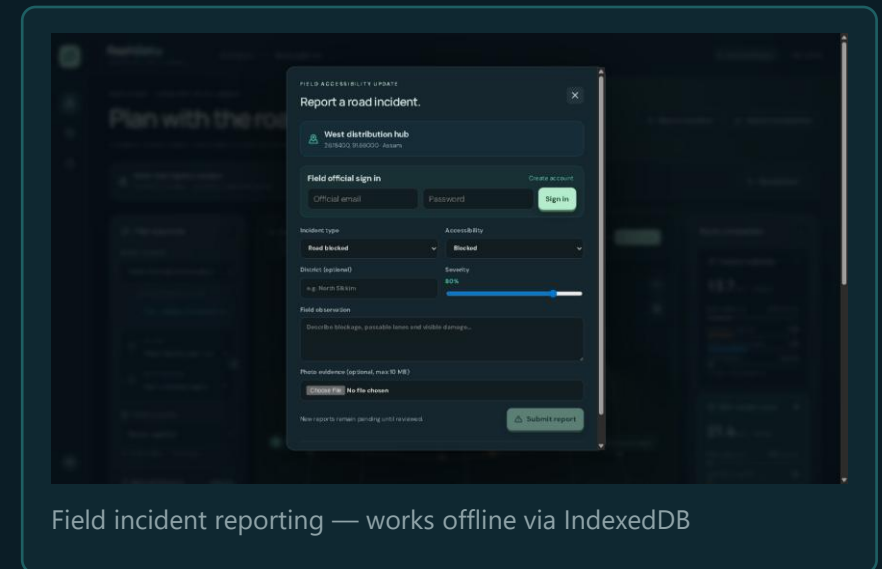
randomized A* vs Dijkstra oracle checks

14/14

live Supabase end-to-end steps verified

1 GB+

public NE data prepared (61 verified sources)



Feasibility and Viability

Not a concept deck — every risk below has a shipped mitigation in the working prototype.

Verified blackspot data for all 8 states is not public

Honest-data architecture: unverified evidence is never auto-applied. Reviewer workflow (field report → candidate roads → accept/reject) ships today; eDAR/MoRTH access is the scaling path.

Full-NE road graph is too large for a free cloud DB

Slim-by-design: 36 MB used of 500 MB free tier. Raw OSM (1.9 GB) stays in versioned files; only metadata, hazards and reviews enter PostGIS.

Rural connectivity is unreliable for field reporting

Installable PWA with offline shell, IndexedDB report queue with automatic retry — verified in tests.

Weather data must stay current

Live Open-Meteo integration (keyless, free) already maps heavy rain/wind onto routing scenarios.

Scaling to a production fleet

State-wise graph loading, Docker + K8s manifests and a 6-person ownership plan are prepared; deployment awaits team VM access.

Impact and Benefits

Essential-supply fleets

Fewer medical/relief consignments trapped on damaged or landslide-affected corridors in the region with India's highest monsoon disruption.

Emergency response

Detour cost known up-front: minutes vs exposure, per vehicle profile — dispatch decisions in seconds, not phone calls.

State transport authorities

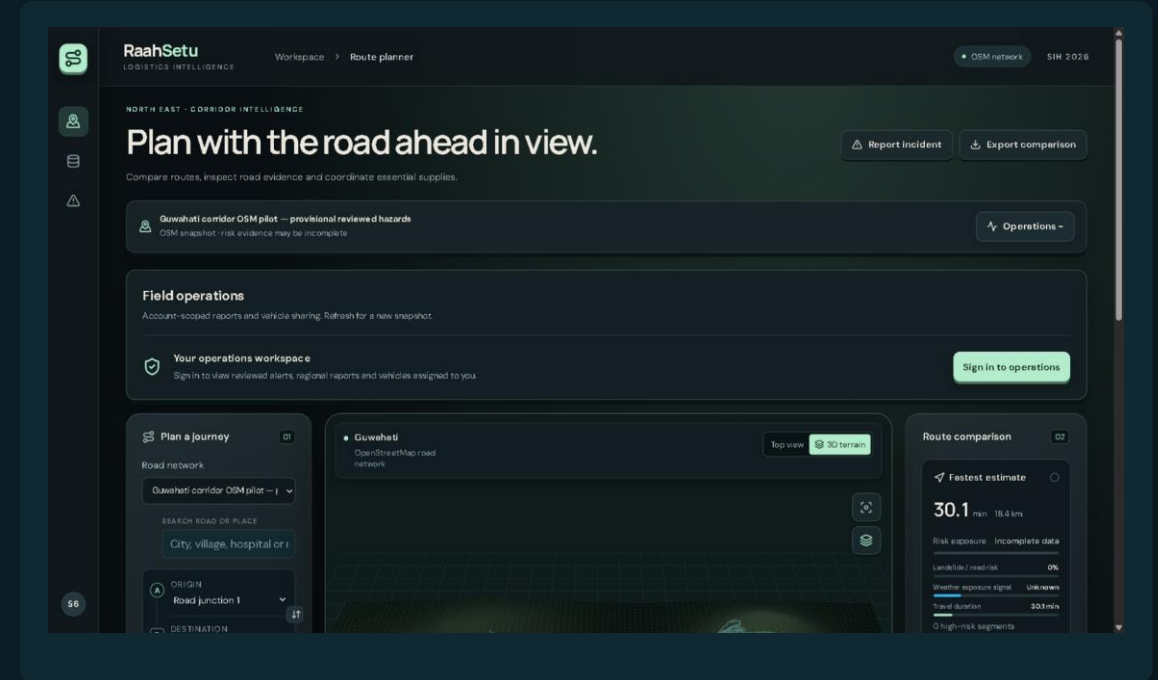
A reviewable, evidence-backed incident pipeline replaces unverified WhatsApp forwards; every closure has provenance.

Drivers on unsafe roads

Explicit high-risk segment warnings instead of silent fastest-route routing.

Runs entirely on free-tier infrastructure today

Supabase free tier (36 MB used) + keyless Open-Meteo + OSM open data — the prototype costs ₹0/month to operate, and scales by adding state graph snapshots one by one.



Fleet operations workspace — GPS, deliveries, alerts on live Supabase

Research and References

OpenStreetMap

North-Eastern zone extract, ODbL 1.0 — road network for all 8 states

openstreetmap.org · geofabrik.de

NASA Global Landslide Catalog

467 historical North-East landslide records used with documented positional accuracy

[.nasa.gov](https://www.nasa.gov) (ArcGIS portal)

Open-Meteo

Live weather API (temperature, precipitation, wind) — keyless, free

open-meteo.com

MoRTH road safety data

State accident statistics consulted; deliberately not mapped to specific roads without verification

morth.nic.in

Supabase / PostGIS

Spatial schema, row-level security, storage for verified evidence

supabase.com

Dijkstra, E. W. (1959)

Independent shortest-path oracle used to verify the custom A* on 300 randomized cases

Numerische Mathematik 1, 269–271

Honest boundaries: risk scores are documented indices, not accident probabilities. Historical hazards are not current closures. Safety claims require reviewed field evidence — and the prototype is built to enforce exactly that.